MILITARY SPECIFICATION

PERSONNEL PLANNING DATA FOR NAVAL SYSTEMS

1. SCOPE

1.1 General. This specification establishes the requirements for Personnel Planning Data (PPD) development during the analysis, design, development, test and evaluation of military weapon and support systems, (less Aviation and Medical). These requirements include:

a. The scope and nature of the work to be accomplished by the contractor to meet the requirements for Personnel Planning Data, and

b. The means by which to ensure the provision of required data.

c. It is the responsibility of the Chief of Naval Operations (OP-01) to direct and coordinate the effort of the Chief of Naval Material and the Chief of Naval Personnel in the effective manpower planning for Navy ships, systems, and equipments. OPNAVINST 5300.3 assigns CNM responsibility for the application of design work study techniques and human engineering, and CNP is assigned responsibility for personnel and training research. Emphasis is placed on the coordination and cooperation of CNM and CNP in the development of enlisted military manpower requirements. This specification is intended to support CNM and CNP in meeting the requirements of CNO in instances in which personnel and training research is conducted under contract.

1.2 Purpose. The purpose of the PPD requirements established herein is to ensure that military systems can be operated, maintained, and supported by the men who constitute military manpower resources by requiring:

a. That at each stage of system development personnel requirements are being assessed in relation to the characteristics imposed by the mission, environment, and hardware of a given system; and

b. That the information needed for effective selection, classification, and training of a sufficient number of system personnel is provided.

The decision as to the applicability of this specification to individual system development projects is the responsibility of the Principal Development Activity in consultation with the Bureau of Naval Personnel.

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids shall form a part of this specification to the extent specified herein:

NUMBER	TITLE
a. NAVPERS 18068	Manual of Qualifications for Advancement in Rating
b. NAVPERS 15105	Navy Enlisted Classifications
c. NAVPERS 15105 (Supplement)	Navy Enlisted Classification Definitions
d. NAVPERS 15839	Manual of Navy Officer Classification
e. NAVPERS 91769	Catalogue of U.S. Naval Training Activities and Courses

2.2 The following documents of the latest issue in effect are listed for information and guidance in complying with the requirements herein:

NUMBER	TITLE
a. SECNAVINST 5430.67	Assignment of Responsibilities for Research, Development, Test and Evaluation
b. OPNAVINST 5300.3	Development and Review of Enlisted Manpower Requirements in Naval Ships
c. BUPERSINST 5300.8	Development and Review of Enlisted Manpower Requirements
d. BUPERSINST 3910.4	Technical Development Plans, Preparation of
e. NAVPERS 18659	Bureau of Naval Personnel New Developments Human Factors Program

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NUMBER	TITLE
f. NAVMATINST 4000.20	Integrated Logistic Support Planning Procedures
g. MIL-H-46855	Human Engineering Requirements for Military Systems, Equipments and Facilities

Copies of applicable documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.

3. REQUIREMENTS

3.1 General. The following requirements for PPD development are applicable to work performed at all stages of military system development. The requirements outlined are designed to assure the effective consideration of personnel resources and requirements during the design process.

3.2 Preliminary PPD Plan. A preliminary PPD Plan shall be submitted as a part of, concurrent with, or separately from the system proposal. This preliminary plan shall include, though need not be limited to, the following:

a. A general statement of scope and objective of the PPD development program;

b. A general plan for meeting requirements which are stated in paragraphs 3.8 through 3.10 including an indication of the methods and documents to be employed;

c. A schedule of PPD development tasks that is related to system development milestones and includes estimated contractor man-hours assigned to each task;

d. An identification and description of the organizational elements responsible for the implementation of this specification and their authority and relationship to management as well as to other system development groups, including any approval relationships established for data flow;

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e. A list of other specifications invoked in this contract which are related to the PPD effort such as MIL-H-46855, the extent to which they are related and how coverage will be acomplished without duplication; and

f. A list of key personnel, by job title and qualifications, to be utilized in the implementation of this specification.

3.3 Data Retention. A PPD file will be established and maintained throughout the life of the contract. This PPD file will contain a record of all situations requiring PPD input, system information used, actions taken and reasons therefore, techniques employed, recommendations made, and the results of the PPD development processes. These data will be available to the Principal Development Activity and the Chief of Naval Personnel as required.

3.4 Nonduplication. Results from the system design and human engineering efforts should be maximally utilized in the accomplishment of PPD development. Extension and transformation of such results provide a means by which to avoid undesirable duplication of effort. The contractor will bring any possible duplication to the immediate attention of the Principal Development Activity. Any instance of undesirable duplication will not be funded.

3.5 Subcontractors. The Principal Development Activity will accept PPD information prepared under this specification on the following conditions:

a. The PPD is developed by the prime contractor's staff, or

b. The PPD is developed by a subcontractor whose personnel have continuing access to all pertinent information throughout the PPD development process.

Any items of this specification that are to be subcontracted shall be identified to the Principal Development Activity. The prime contractor shall be responsible for integration and the availability of information necessary to meet the requirements of this specification.

3.6 PPD Reports. Table 1 presents a summary of report requirements.

a. The PPD Reports shall be submitted through the Principal Development Activity to the Chief of Naval Personnel. The PPD Reports shall in each instance reflect the most recent system design configuration and shall include the information categories indicated in paragraph 3.8.

b. PPD Progress reports shall be submitted quarterly. These progress reports shall summarize progress made during the period and

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milestones accomplished, identify problem areas, specify changes in personnel, and summarize man-hours applied during the period.

TABLE 1

SUMMARY OF REPORT REQUIREMENTS

PPD REPORT REQUIRED	Spec. Para.	Department of the Navy Action	When Due
Preliminary PPD Plan	3.2	Evaluation	Concurrent with proposal
PPD Reports	3.6	Review and approval	As specified in the contract
PPD Progress Reports	3.6	Review	Quarterly
Revised PPD Plan	3.7	Revision of PPD Plan at guidance meeting	30 days after guidance meeting
Working papers	3.8	Review and guidance	As required
Studies requirements	3.9	Review and disposition	Concurrent with PPD Data Reports and as needed
PPD test plan	3.10	Review and approval	30 days prior to prototype tests
PPD test results	3.10	Review and analysis	30 days following completion of prototype tests

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3.7 Coordination. A meeting will be arranged among representatives of the contractor, the Principal Development Activity, and the Chief of Naval Personnel not later than 30 days after contract award. The Preliminary PPD Plan as submitted with the contractor's proposal and/or as modified as a result of contract negotiation will be reviewed, discussed and revised at this time. In addition, the results of any available previous personnel research conducted for the system will be provided, and an agreed upon number of briefing sessions will be scheduled to acquaint contractor personnel with Navy manpower capabilities and resources. Formal coordination and guidance meetings will be held as needed but at least at 90-day intervals to coincide with Progress Reports, and will provide continuing guidance concerning PPD development. Informal contact will proceed on a continuing basis. The contractor must permit reasonable inspection of ongoing efforts which relate to the requirements of this specification.

3.8 Personnel Planning Data. Personnel Planning Data will be developed to provide the basic information required for the PPD Reports. Analyses will be conducted to a level of detail consistent with the systems progress in the RDT&E cycle and will be updated as appropriate to reflect the actual system configuration and status. By examination and utilization of available documentation or by direct development, the information categorized in the following listing will be documented by the contractor.

3.8.1 System Description. The contractor will provide descriptions of the system's mission, function, equipment, implementation, and operation and maintenance philosophy in order to provide the system context within which the PPD is developed.

a. Mission Description

A summary mission description reflecting the strategic, tactical and nontactical objectives of the system and its operational requirements will be provided. The contractor shall describe the maximum simultaneous external load that the system will be required to handle. Appropriate mission profiles with time line, quantitative performance objectives, and environmental factors having personnel implications will be provided.

b. Functional Description

A description of system and subsystem functions, and diagramatic depiction of system and subsystem interactions and sequences.

c. Detailed Equipment Descriptions

A brief narrative description of the function of each equipment; the equipment design configuration such as workspace layout, console design, panel layout, etc., supported by illustrations and justification

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of the design features of the equipment in as far as they relate to the quantity and quality of persons required for system maintenance and operation.

d. Implementation Description

An installation plan and schedule describing location of units, number of units, number of installations per unit, date of commencing installation, expected operational date of equipment, and system(s) replaced (if any). The current manpower level and structure of the unit being replaced shall be described.

e. Operation and Maintenance Description

This description will include equipment operation and maintenance philosophy and assumptions as relevant to personnel. The contractor will provide a delineation of the system operation and maintenance levels, the philosophy for each level, and the maintainability and operability tolerances to meet system requirements as stated in the system guidance documents. Design efforts to meet current availability and capability of Naval Personnel shall be indicated.

3.8.2 Man-Machine Allocation. The allocation of functions between men and machines shall be reviewed by the contractor. The review will result in either justification and concurrence with the allocations and/or an identification of questionable allocations or problem areas. Studies of alternative allocations which will enhance system effectiveness will be recommended. A schematic depiction of man-machine assignments shall be provided in addition to the narrative description.

3.8.3 Task Analysis. Operator and maintenance task data will be separately developed, within the availability of sufficient operational and equipment information, to justify personnel requirements. All system functions involving personnel shall be analyzed to identify tasks requiring interaction of personnel with equipment and other personnel. Identification of tasks shall commence as soon as possible after the establishment of the overall system concept and functions in order to permit their consideration in design. The specificity of the analysis shall be commensurate with the level of design detail. The level of analysis should provide sufficient detail to justify the quantitative and qualitative personnel requirements recommended for the system. The task analysis shall include: system operation, operator maintenance, and each echelon of maintenance up to and including depot maintenace. The analysis shall be conducted to provide the following information as appropriate and as justified by the contractor in the preliminary plan.

a. Task identification. Each task shall be given a separate title with numeric or alpha-numeric designation. Tasks identified shall be grouped by subsystem and equipment in temporal order.

b. Equipment Status. Description of system/equipment condition prior to the task accomplishment.

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c. Environment and location. The task location shall be given. The environment in which the task must be performed shall be described.

d. Criticality of Task Performance. Effect on system if task were not performed satisfactorily.

e. Frequency. Frequency of task recurrences based on system/ equipment operating time and maintenance requirements.

f. Task Time. Man-minutes required to complete a task. (Continuous and time-shared tasks should be so identified).

g. Task Schedule. Identify tasks to be accomplished on a scheduled or unscheduled basis.

h. Safety. Precautions that must be taken to assure safe performance of task.

i. Special Requirements. Identify and describe tasks which may require emphasis due to unusual requirements for speed of action, perception, coordination, recall of information, or physical dexterity.

j. Equipment. The tools, equipment, and materials required to accomplish the task shall be listed and identified.

k. Initiation Cue. Task initiation stimulus shall be described to clarify the requirement.

l. Completion Cue. A description of the feedback which indicates the task completion.

m. Personnel. The number of personnel required to accomplish the particular task.

n. Personnel performance standards. Based on mission requirements, performance standards shall be identified for all tasks in order to meet system performance requirements and to optimize overall system effectiveness. Standards shall specify requirements for speed of performance, and tolerance limits for acceptable performance.

o. Skill and knowledge requirement. The skills and knowledges required to perform will be identified for each task. Both skill area and level of skill shall be identified.

p. Emergency Procedures. Emergency procedures shall be analyzed in detail to permit preparation of comprehensive and accurate operation and maintenance procedures. Where a complete analysis of emergency tasks is not feasible, owing to their volume or unpredictable nature, a representative sample of critical emergency tasks shall be analyzed to ensure accurate identification of knowledges, aptitudes, physical characteristics,

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and other measurable personnel attributes required to permit personnel to handle all such tasks successfully.

3.8.4 Position Identification. Individual task sequences will be organized into jobs or positions. Tasks and task elements will be organized into positions in such manner as to achieve:

a. Fullest utilization of available work time.

b. Maximum possible utilization of performance capabilities already existing in operational units.

c. The assignment of tasks to positions with minimum task time conflicts.

d. Performance of all tasks of a single individual without excessive movement, travel time, etc.

e. Homogeneity of required capabilities for each billet to facilitate selection and training.

f. Delineation of jobs to conform as closely as possible to the existing Navy job classification scheme. Guidance may be obtained from NAVPERS 18068, NAVPERS 15105 and NAVPERS 15839.

g. Clear separation of required abilities in order to provide effective utilization of personnel representing differing levels of responsibility, aptitude, training, and/or experience.

h. Special consideration shall be given to the identification of positions which can utilize personnel with low skill level whenever consistent with effective system operation to permit maximum utilization of short-term or low-aptitude personnel. Care should be taken to ensure that:

1. System performance is not jeopardized;

2. Necessary supervision or assistance can be obtained from higher level personnel without interference in the primary work responsibilities of such personnel;

3. Where unexpected task demands are encountered the tasks exceeding capabilities of lower level personnel can be identified rapidly and referred to higher level personnel without delay or jeopardy to system performance and personnel safety.

3.8.5 Knowledge and Skill Requirements. The knowledge and skill requirements for each operator and maintenance position shall be identified.

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The guidance contained in the current editions of NAVPERS 18068, 15839 and 15105 will be used in relating knowledge and skill requirements data for system operation and maintenance jobs to grades, ratings and rates. The guidance contained in BUPERS INST 3910.4B will be used to list the new knowledge and skill requirements from among the total listed.

3.8.6 Job Summary. A summary will be provided for each job or position required for operation and maintenance. Such summary will indicate the position title, grade level, recommended personnel classifications code (new codes may be indicated), duty station, narrative description of job, task requirements (may be referenced to task analysis), special selection requirements, training requirements (pre-requisites, applicable existing courses and/or new skills and knowledges to be identified).

3.8.7 Quantitative Requirements. Sufficient numbers of personnel will be recommended to ensure completion of required operations under conditions Steaming Operational (Condition III) and General Quarters (Condition I). Recommended manning levels will be based upon consideration of:

a. Frequency and duration of scheduled tasks;

b. Estimated frequency and duration of unscheduled (contingency) tasks;

c. System performance requirements, e.g., equipment availability, turn-around time, etc.;

d. Time during which system will be manned, e.g., work shifts;

e. Anticipated loss of personnel time due to travel, sick leave, etc.;

f. Logical personnel substitution within the task complex for emergency situations;

g. Space constraints;

h. Available skill levels; and

i. Manpower availability.

Data will be developed to meet the requirements outlined in BUPERS INST 3910.4B.

3.8.8 Training Requirements. The Bureau of Naval Personnel is responsible for the design and development of Bureau of Naval Personnel school programs. The contractor will, however, recommend for consideration therein the training necessary for system operation and maintenance. Special needs

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for training devices and for equipment will also be identified. These needs shall be communicated to the Naval Training Devices Center which is responsible for recommending training devices and simulation requirements in support of the training necessary for system operation and maintenance. The data indicated by BUPERS INST 3910.4B will be developed using the guidance contained in the current edition of NAVPERS 91769.

3.8.9 Manpower Summary. Summaries will be made of required number of personnel for each position, classification code, system and subsystem, and subtotals and totals provided. A summary will be made of the manpower requirements and recommended organizational units within the system. Such a summary will include the quantitative requirements for personnel to man the system under all modes of operation.

3.8.10 Provision of Data. The contractor shall make available to the Principal Development Activity the results, in working paper form, of each of the analyses outlined in paragraph 3.8 as they are completed. These working paper submissions will facilitate continuing review of the progress and applicability of PPD development by the Chief of Naval Personnel and will provide a substantive basis for continuing guidance.

3.9 Personnel and Training Studies Requirements. Requirements for special PPD studies will be submitted to the Principal Development Activity for use by the Chief of Naval Personnel. Particular attention should be given to the requirements for studies and/or dynamic simulations of critical system performance. A research plan for each study to include equipment/facilities, subjects, and estimated man-hours required, and other costs should be submitted to the Principal Development Activity for disposition.

3.10 PPD in Contractor Test of Prototype Equipment.

3.10.1 At a coordination meeting scheduled prior to beginning contractor tests of the prototype equipment, the contractor will submit to the Principal Development Activity a PPD test plan to be accomplished as an integral Activity portion of the prototype evaluation which will be coordinated with other tests. The objective of the PPD test during prototype evaluation is to confirm and validate recommended manning. This PPD portion of the test plan shall include but not be limited to the following:

a. PPD Experimental Design. The detailed plan that will permit the collection of reliable, valid, and meaningful data with regard to the parameters under test.

b. Test Population Parameters. To include composition of the test operator and maintenance team, previous experience and training, and degree of similarity to Naval personnel capabilities.

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c. Simulation Parameters. To include requirements for simulation, special computer programs or routines, degree of realism, and degree of simulation of system load and environment, and others as required.

d. Equipment, Facilities, and Job Aids to be Employed.

e. Personnel Performance Standards. Based on detailed task description and analysis data.

f. Specific Tasks to be Tested. To include all tasks that are considered critical with respect to speed, accuracy or reliability of system performance and a representative sample of noncritical scheduled and unscheduled operator and maintenance tasks.

g. Listing of Data to be Collected.

h. Methods and Procedures for Testing.

i. Data-Collection Forms. To include any requirements for supporting computer programs or routines for data-recording, reduction, and formatting purposes.

3.10.2 During the conduct of the contractor's prototype tests provision will be made to accommodate representatives of the Principal Development Activity and the Chief of Naval Personnel who will observe the conduct of the tests and conduct follow-up interviews with personnel assigned to operations and maintenance positions. Following completion of prototype tests the PPD portions of test results will be made available to the Principal Development Activity for coordination with the Chief of Naval Personnel.

4. QUALITY ASSURANCE

Compliance with the requirements of this specification as contractually agreed upon will constitute the criterion for acceptance of the contractor effort required by this specification.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

This section is not applicable to this specification.

Preparing Activity: Chief of Naval Material

Project Number:
MISC - N525

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